PIPELINE AND HAZARDOUS MATERIALS SAFETY ADMINISTRATION
U.S. DEPARTMENT OF TRANSPORTATION

Docket No. PHMSA-2019-0091

Comments of Earthjustice and Environmental Defense Fund on *Pipeline Safety: Amendments to Liquefied Natural Gas Facilities*, 90 Fed. Reg. 18,949 (May 5, 2025) and *Pipeline Safety: Notice of Intent to Prepare an Environmental Impact Statement for the Amendments to Liquefied Natural Gas Facilities Rulemaking*, 90 Fed. Reg. 24,088 (June 6, 2025)

Submitted July 7, 2025

Earthjustice and Environmental Defense Fund (EDF) submit these comments in response to the Pipeline and Hazardous Materials Safety Administration's (PHMSA) advance notice of proposed rulemaking (ANPRM) and notice of intent to prepare an environmental impact statement for planned updates to its regulations for liquefied natural gas (LNG) facilities.

We support PHMSA's proposal to update the regulations to incorporate the newest 2023 version of National Fire Protection Association (NFPA) standard 59A, in place of the outdated 2001 version currently incorporated, but making this update would not fulfill either (1) PHMSA's statutory duty to issue regulations adequate to protect public safety and the environment or (2) the specific mandates in the PIPES Act of 2016, Pub. L. No. 114-183, § 27(b), 130 Stat. 514, and PIPES Act of 2020, Pub. L. No. 116-260, § 110(a), 134 Stat. 1182, for PHMSA to update its regulations for small-scale and large-scale LNG facilities. While PHMSA may be able to act quickly to incorporate the new version of NFPA 59A, it should ensure it considers all relevant issues when adopting comprehensive safety standards for LNG facilities in a future rulemaking. The information solicited in the ANPRM is also not sufficient to inform a comprehensive update of PHMSA's LNG standards, as the ANPRM does not address all statutorily-required factors, such as safety and environmental information and safety and environmental costs and benefits, but instead focuses primarily on costs and benefits to industry.

The ANPRM also does not address several topics which must be considered as part of a comprehensive update to PHMSA's LNG standards. First, it does not fully address the unique risks posed by the liquefaction facilities present at export terminals, none of which even existed in the U.S. at the time of the last major amendments to PHMSA's LNG standards. Second, it does not incorporate lessons learned from safety incidents at LNG facilities. Third, it does not address the most updated science and engineering or previously-identified knowledge and regulatory gaps. PHMSA must collect information on these topics and solicit new research as necessary to inform its LNG rulemaking.

We further urge PHMSA to provide adequate opportunity for public engagement, including a public hearing in the Gulf communities most impacted by the LNG export industry.

We also provide responses to several of the specific topics in the ANPRM. PHMSA should clarify that reporting of *all* safety-related conditions is required (Topic C.3). Adopting the 2023 version of NFPA 59A is a reasonable interim step while PHMSA works toward a comprehensive LNG rulemaking (Topic D.2). The recent executive orders directing agencies to adopt a deregulatory approach do not alter PHMSA's statutory duties, which include: to issue LNG regulations sufficient to protect public safety and the environment; to update its rules for small-scale and large-scale LNG facilities in the manner specifically directed by Congress; to consider environmental and safety information, among other required considerations, in promulgating rulemakings; to consider costs and benefits, including environmental and safety costs and benefits; and to only issue rulemakings whose benefits outweigh their costs (Topic H).

Finally, we provide scoping comments to inform PHMSA's preparation of an environmental impact statement on the proposed rulemaking in compliance with the National Environmental Policy Act (NEPA).

CONTENTS

I. PHMSA has a legal obligation to update its LNG safety rules to better promote safety and protect the environment.

PHMSA should ensure that any LNG rulemaking it issues fulfills PHMSA's core statutory directive regarding LNG facilities: to regulate them in a manner that will promote safety and protect the environment, which includes updating the regulations when technology or the industry has changed such that the regulations are no longer effectively minimizing risk to the public and workers and harms to the environment. 49 U.S.C. § 60102(a)(2), (b)(1)–(3); *id.* § 60103(a), (b), (d). In addition, PHMSA must fulfill the specific statutory mandates enacted by Congress to update its LNG safety rules for small-scale LNG facilities[1] and large-scale LNG facilities.[2]

The questions that PHMSA poses in its ANPRM—which largely pertain to costs to industry and to incorporating a newer version of voluntary industry standard NFPA 59A—do not, however, solicit sufficient information necessary for PHMSA to promulgate a rulemaking that would fulfill either its core statutory obligations or more recent Congressional mandates to update its LNG safety regulations. PHMSA would need to collect significantly more information, in addition to information on the topics listed in the ANPRM, to inform a comprehensive rulemaking that fulfills its statutory mandates.

Adopting the 2023 version of NFPA 59A will not fulfill PHMSA's outstanding mandates relating to LNG facilities, though updating to the newest version of NFPA 59A is a reasonable interim step while PHMSA works toward a more comprehensive update to its LNG facilities

---

[1] PIPES Act of 2016, § 27(b), Pub. L. No. 114-183, June 22, 2016, 130 Stat 514.
[2] PIPES Act of 2020, § 110, Pub. L. 116-260, December 27, 2020, 134 Stat 1182.

standards. Finally, PHMSA's contemplated "deregulatory" approach is not consistent with fulfilling its statutory mandates.

A. PHMSA has statutory mandates to set LNG standards sufficient to protect safety and the environment and to specifically update its rules for small-scale and large-scale LNG facilities.

The Pipeline Safety Act requires PHMSA to set "minimum safety standards" for pipeline transportation and facilities that are "practicable" and "designed to meet the need for … safety….and…protecting the environment." 49 U.S.C. § 60102(a)(2), (b)(1). In setting any standards, PHMSA must consider "relevant available…gas pipeline safety information…hazardous liquid pipeline safety information…and…environmental information." *Id.* § 60102(b)(2).

Regarding LNG facilities, the Pipeline Safety Act directs PHMSA to set: "minimum safety standards for deciding on the location of a new liquefied natural gas pipeline facility," § 60103(a); "minimum safety standards for designing, installing, constructing, initially inspecting, and initially testing a new liquefied natural gas pipeline facility," § 60103(b); and "minimum operating and maintenance standards for a liquefied natural gas pipeline facility," § 60103(d). The statute lists the specific factors that PHMSA must consider when setting those standards. In setting location standards, PHMSA must consider the:

> (1) kind and use of the facility;
>
> (2) existing and projected population and demographic characteristics of the location;
>
> (3) existing and proposed land use near the location;
>
> (4) natural physical aspects of the location;
>
> (5) medical, law enforcement, and fire prevention capabilities near the location that can cope with a risk caused by the facility;
>
> (6) need to encourage remote siting; and

> (7) national security.

§ 60103(a). When issuing design, installation, construction, inspection, and testing standards, PHMSA must consider:

> (1) the characteristics of material to be used in constructing the facility and of alternative material;
>
> (2) design factors;
>
> (3) the characteristics of the liquefied natural gas to be stored or converted at, or transported by, the facility; and
>
> (4) the public safety factors of the design and of alternative designs, particularly the ability to prevent and contain a liquefied natural gas spill.

§ 60103(b). The factors PHMSA must consider in setting operation and maintenance standards are:

> (1) the conditions, features, and type of equipment and structures that make up or are used in connection with the facility;
>
> (2) the fire prevention and containment equipment at the facility;
>
> (3) security measures to prevent an intentional act that could cause a liquefied natural gas accident;
>
> (4) maintenance procedures and equipment;
>
> (5) the training of personnel in matters specified by this subsection; and
>
> (6) other factors and conditions related to the safe handling of liquefied natural gas.

§ 60103(d).

In adopting any standards, PHMSA must "propose or issue a standard . . . only upon a reasoned determination that the benefits, including safety and environmental benefits, of the intended standard justify its costs." 49 U.S.C. § 60102(b)(5). Thus, the agency's required risk assessment that must "identify the costs and benefits associated with the proposed standard" should address safety and environmental benefits. *See* 49 U.S.C. § 60102(b)(3).

The ANPRM inappropriately fails to solicit relevant available safety or environmental information as required by § 60102(b)(2) or information relevant to the enumerated factors in § 60103 that PHMSA ***must*** consider when amending its LNG safety standards. Instead, the ANPRM narrowly focuses on costs and regulatory burdens. The statute does not list regulatory burden or cost to industry as a consideration factor for the LNG safety standards, apart from the general requirements to identify costs and benefits that apply to all PHMSA's pipeline and LNG rulemakings. 49 U.S.C. § 60102(b)(2)(D), (E).

Because PHMSA can adopt standards "only upon a reasoned determination that the benefits, including safety and environmental benefits, of the intended standard justify its costs," § 60102(b)(5), a rulemaking process focused solely on the costs or benefits to industry, without consideration of safety or environmental costs and benefits to the public, is inadequate.

As PHMSA recognizes in the ANPRM, *Pipeline Safety: Amendments to Liquefied Natural Gas Facilities,* 90 Fed. Reg. 18,949, 18,950 (May 5, 2025), the agency currently has two unfulfilled statutory mandates to update its LNG facilities rules. In the PIPES Act of 2016, Congress directed PHMSA to "review and update the minimum safety standards prescribed pursuant to section 60103 of title 49, United States Code, for permanent, small scale liquefied natural gas pipeline facilities." *Id.* § 27(b). In the PIPES Act of 2020, Congress directed PHMSA to review its Part 193 LNG safety rules and to update the regulations applicable to "large-scale liquified natural gas facilities (other than peak shaving facilities) to provide for a risk-based regulatory approach." *Id.* § 110(a). Congress required that the updated LNG safety regulations must "achieve a level of safety that is equivalent to, or greater than, the level of safety required by the standards prescribed as of the date of enactment of this Act." *Id.* § 110(b)(2). In the PIPES Act of 2020 Section 110, Congress also gave specific instructions for the contents of the regulations, directing PHMSA that these updated safety regulations must, “at a minimum,” require operators of LNG facilities:

> (1) to develop and maintain written safety information identifying hazards associated with—
>
> > (A) the processes of liquefied natural gas conversion, storage, and transport;
> >
> > (B) equipment used in the processes; and
> >
> > (C) technology used in the processes;
>
> (2) to conduct a hazard assessment, including the identification of potential sources of accidental releases;
>
> (3)(A) to consult with employees and representatives of employees on the development and execution of hazard assessments under paragraph (2); and
>
> > (B) to provide employees access to the records of the hazard assessments and any other records required under the updated standards;
>
> (4) to establish a system to respond to the findings of a hazard assessment conducted under paragraph (2) that addresses prevention, mitigation, and emergency responses;
>
> (5) to review, when a design change occurs, the most recent hazard assessment conducted under paragraph (2) and the response system established under paragraph (4);
>
> (6) to develop and implement written operating procedures for the processes of liquefied natural gas conversion, storage, and transport;
>
> (7)(A) to provide written safety and operating information to employees; and
>
> > (B) to train employees in operating procedures with an emphasis on addressing hazards and using safe practices;
>
> (8) to ensure contractors and contract employees are provided appropriate information and training;
>
> (9) to train and educate employees and contractors in emergency response;
>
> (10) to establish a quality assurance program to ensure that equipment, maintenance materials, and spare parts relating to the operations and maintenance of liquefied natural gas facilities are fabricated and installed consistent with design specifications;

> (11) to establish maintenance systems for critical process-related equipment, including written procedures, employee training, appropriate inspections, and testing of that equipment to ensure ongoing mechanical integrity;
>
> (12) to conduct pre-start-up safety reviews of all newly installed or modified equipment;
>
> (13) to establish and implement written procedures to manage change to processes of liquefied natural gas conversion, storage, and transport, technology, equipment, and facilities; and
>
> (14)(A) to investigate each incident that results in, or could have resulted in—
>
> (i) loss of life;
>
> (ii) destruction of private property; or
>
> (iii) a major accident; and
>
> (B) to have operating personnel—
>
> (i) review any findings of an investigation under subparagraph (A); and
>
> (ii) if appropriate, take responsive measures.

Id. § 110(c). PHMSA has long-since missed the PIPES Act of 2020's deadline to issue a final rule by December 27, 2023. *Id.* § 110(a).

B. The information solicited in the ANPRM is not sufficient to inform a rulemaking that would fulfill PHMSA's statutory mandates.

In order to fulfill these mandates, PHMSA must collect information related to the risks that LNG facilities currently pose to their communities and workers and the ways in which these risks can be best reduced or eliminated, as well as information about the environmental harms posed by the LNG facilities and the ways in which these harms can be mitigated or eliminated. PHMSA then needs to craft regulations that implement practices that will most effectively reduce those hazards and protect the environment, including practices that may not be currently used by any operational LNG facilities. Many of the specific questions posed in the ANPRM, however, relate not to such safety or environmental information but to the costs and benefits to industry of making various small regulatory changes, including updating the NFPA 59A standards that are incorporated at 49 C.F.R. § 193.2051 (LNG facility siting), § 193.2101(a) (design), § 193.2301 (construction), and § 193.2401 (equipment) from the 2001 to the 2023 version. *See* 90 Fed. Reg. at 18,951–53, Topics A through H.

Merely adopting the updated 2023 version of NFPA 59A—a long-overdue and necessary update—would not fulfill PHMSA's statutory duty to regulate LNG facilities to promote safety and environmental protection or its two outstanding LNG rulemaking mandates. PHMSA seems to acknowledge this fact by suggesting that it may initiate one rulemaking to update the version of the incorporated NFPA 59A standard and then, in the future, undertake a separate rulemaking to fulfill its outstanding statutory mandate. 90 Fed. Reg. at 18,951 (Topic D.2).

Taking a "deregulatory" approach to this rulemaking—as PHMSA suggests it may, 90 Fed. Reg. at 18,953, Topic H.4—would also be inconsistent with the agency's statutory duty and with the two specific LNG rulemaking mandates imposed by Congress. The PIPES Act of 2020 specifically requires that PHMSA's updated risk-based standards for large scale LNG facilities must "achieve a level of safety that is equivalent to, or greater than, the level of safety required by the standards prescribed as of the date of enactment of this Act." § 110(b)(2). Any deregulatory actions that reduce the protectiveness of safety standards are contrary to that directive. Additionally, as explained further below, *see* response to Topic H.4, an executive order directing agencies to take a deregulatory approach and to reduce burdens to industry cannot overrule statutes passed by Congress, including the two statutes that mandate PHMSA make specific updates to its LNG safety rules for small scale and large-scale LNG facilities.

II. PHMSA must update its LNG safety standards based on the best available science and engineering to address the risks and impacts of the expanded export LNG industry.

There is an urgent need for PHMSA to update its LNG safety standards, which were drafted at a time when the only large LNG terminals in the country were import facilities. The changes contemplated by the ANPRM fall short of the necessary update because it focuses largely on the costs and benefits to industry and fails to consider the appropriate scope and contents of updated safety regulations for LNG facilities. For example, PHMSA should take into account the current and anticipated future LNG facilities and tailor its oversight to minimize the risk posed by large LNG export terminals. The agency acknowledges that current regulations are out of date and were drafted before the export industry rapidly expanded or the U.S. had any large export terminals, both of which pose unique safety and environmental risks. Similarly, significant safety incidents have occurred at LNG facilities since the last rulemaking, and PHMSA must now incorporate lessons learned from those incidents. Further, researchers and participants in PHMSA R&D workshops have identified numerous knowledge gaps and regulatory gaps, and PHMSA's rulemaking must address both kinds of gaps.

A. The current regulations are out of date and must be updated to protect safety and the environment in light of the rapid LNG export industry expansion, new risks, and new technology and technical standards.

PHMSA's LNG regulations are out of date and must be updated to account for significant changes in technology and in the industry that have occurred since they were last amended. The first U.S. LNG export terminal came online in February 2016.[3] Since that time, the U.S. has

[3] U.S. Department of Energy, *U.S. Liquefied Natural Gas (LNG) Exports Fact Sheet* (Mar. 2025), https://www.energy.gov/sites/default/files/2025-03/U.S.%20Liquefied%20Natural%20Gas%20%28LNG%29%20Exports%20Fact%20Sheet_0.pdf; Scott DiSavino, *After six decades, U.S. set to turn natgas exporter amid LNG boom,* Reuters (Mar. 29, 2017), https://www.reuters.com/article/markets/stocks/after-six-decades-us-set-to-turn-natgas-exporter-amid-lng-boom-idUSKBN1700F0/.

rapidly expanded its export capacity, annually processing and exporting more LNG than any other country since 2023.[4] In 2024, the U.S. exported 11.9 billion cubic feet per day of LNG.[5] As PHMSA acknowledges, however, the last significant amendments to PHMSA's LNG regulations occurred in 2004, long before this industry transition occurred. 90 Fed. Reg. at 18,950 (citing *Pipeline Safety: Liquefied Natural Gas Facilities; Clarifying and Updating Safety Standards*, 69 Fed. Reg. 11,336 (Mar. 10, 2004)).

One of the necessary amendments to PHMSA's LNG regulations is to update the technical standards that are incorporated by reference in the regulations. The Government Accountability Office (GAO) found in a 2020 report that eight of the nine technical standards incorporated in PHMSA's regulations for LNG facilities were outdated at that time.[6] GAO issued two recommendations to PHMSA as a result of that report, both of which remain outstanding: to "conduct a standards-specific review of regulations that incorporate standards and, if necessary, update the regulations or document its decision for not updating them"; and to "establish a process to conduct a standards-specific review of regulations that incorporate standards every 3 to 5 years and to update the regulations, if necessary."[7]

In this rulemaking, PHMSA is proposing to update one of those outdated technical standards—the 2001 version of National Fire Protection Association (NFPA) standard 59A—to the most recent 2023 version of the standard. Updating the regulations to incorporate the newest version of this standard could create significant safety benefits, as well as creating cost-savings

---

[4] EIA, *The United States was the world's largest liquefied natural gas exporter in 2023* (Apr. 1, 2024), https://www.eia.gov/todayinenergy/detail.php?id=61683; EIA, *The United States remained the world's largest liquefied natural gas exporter in 2024* (Mar. 27, 2025), https://www.eia.gov/todayinenergy/detail.php?id=64844.

[5] EIA, *The United States remained the world's largest liquefied natural gas exporter in 2024*, *supra* note 4.

[6] GAO, GAO–20–619, *Natural Gas Exports: Updated Guidance and Regulations Could Improve Facility Permitting Processes* at 29–30 (Aug. 2020), https://www.gao.gov/assets/gao-20-619.pdf.

[7] *Id.* at 35.

for industry, as GAO's 2020 report found that facility operators were incurring additional costs to show that equipment certified to newer technical standards could also meet the older, outdated standards.[8] Indeed, multiple industry rulemaking petitions have requested PHMSA to incorporate the newer version of NFPA 59A.

Merely updating the technical standards that are incorporated, however, is not sufficient to ensure that PHMSA's safety regulations are appropriately mitigating the risks from an industry that looks very different today than it did when the current standards were drafted.

The regulations must account for the different safety risks export terminals pose as compared to import terminals or small-scale peak shaving facilities. The liquefied methane that is present at all LNG facilities poses its own safety hazards, such as pool fires,[9] which can burn at temperatures much higher than those of oil fires.[10] Because a vapor cloud of methane that has boiled off from LNG is flammable at concentrations of 5 to 15%,[11] even a small leak can cause a fire. Moreover, the liquefaction equipment at export terminals requires the presence of other hydrocarbons and refrigerants, which pose their own safety hazards, including increased risk of vapor cloud explosions. *See, infra*, Section II.C.

i. LNG facilities are geographically concentrated and contribute to cumulative impacts for nearby communities.

PHMSA should also update the regulations to account for the cumulative risks faced by the communities located in close proximity to LNG export facilities. Six of the eight currently-operational LNG export facilities in the United States are along the Gulf Coast in Texas and

[8] *Id.* at 29–30.

[9] *Id.* at 2.

[10] GAO, GAO-07-316, *Public Safety Consequences of a Terrorist Attack on a Tanker Carrying Liquefied Natural Gas Need Clarification*, at 9 (2007), https://www.gao.gov/assets/gao-07-316.pdf.

[11] *Id.* at 8.

Louisiana.[12] Many of the additional export terminals that have been approved by the Federal Energy Regulatory Commission but are not yet operational (some of which are already under construction) are located along the Gulf Coast: seven in Texas, six in Louisiana, one in Mississippi, and one in the Gulf offshore of Southwest Louisiana.[13] Within these regional clusters, both operational and planned export terminals operated by separate developers will be located in close proximity. For example, Texas LNG Brownsville (planned) and Rio Grande LNG Phase I (under construction) and Phase II (planned) are all located along a few miles of the same waterway.[14] Sabine Pass LNG Phases I & II (operating), Sabine Pass Phase V (planned), Port Arthur LNG (under construction), Port Arthur LNG Expansion (planned), and Golden Pass LNG (under construction) are all located within a few miles of each other along Sabine Pass, which forms the border between southern Texas and Louisiana.[15] Calcasieu Pass LNG (operating), CP2 LNG Phases I & II (planned), and Commonwealth LNG (planned) are all located at Calcasieu Pass at the mouth of the Calcasieu River.[16] Further up the Calcasieu River, on the shipping channel and at the north end of Calcasieu Lake are: Cameron LNG Phase I (operating); Cameron LNG Phase II (planned); Woodside LNG Phase I (under construction) and Woodside LNG Phase II (planned); Magnolia LNG (planned); and Lake Charles LNG (planned). Magnolia LNG and Lake Charles LNG are proposed to sit directly across from each other, on a canal just off the River, with Woodside LNG Phases I and II nearby at the mouth of that same

[12] EIA, *The eighth U.S. liquefied natural gas export terminal, Plaquemines LNG, ships first cargo* (Jan. 13. 2025), https://www.eia.gov/todayinenergy/detail.php?id=64224.

[13] FERC, *U.S. LNG Export Terminals – Existing, Approved not Yet Built, and Proposed* (June 17, 2025), https://www.ferc.gov/media/us-lng-export-terminals-existing-approved-not-yet-built-and-proposed. Note that FERC lists Venture Global's Plaquemines LNG facility in Southeast Louisiana as "under construction," but it has, in fact, commenced shipments of LNG. *See* EIA, *supra* note 12.

[14] *U.S. LNG Export Tracker*, Sierra Club, https://www.sierraclub.org/dirty-fuels/us-lng-export-tracker.

[15] *Id*.

[16] *Id*.

canal.[17] These facilities each include multiple liquefaction trains, and many of the operational and under-construction facilities have plans to expand by adding more liquefactions trains, including: Corpus Christi Stage III and Corpus Christi Trains 8–9; CP2; Plaquemines LNG Expansion; Sabine Pass Stage 5; and Cameron Train 4.[18]

This high concentration of separate facilities, as well as expansion of facilities to comprise even more co-located liquefaction trains, creates cumulative safety risks to communities and increases the risk of cascading failures. Many LNG export terminals are also located near other industrial infrastructure that may increase cumulative safety risks for nearby communities. For example, there are many petrochemical facilities and oil refineries along the Gulf Coast; the City of Port Isabel, Texas, is near one planned LNG export terminal and one existing LNG export terminal (with a planned expansion) and the SpaceX Starbase launch facility[19]; and many LNG facilities or nearby petrochemical plants are planning to install carbon capture equipment, which comes with the risk posed by pipelines carrying carbon dioxide to locations at which it can be sequestered, or of onsite carbon sequestration.[20] The cumulative safety risks associated with these concentrated LNG export facilities are in addition to their cumulative impacts to waterways—both from wastewater discharges and from LNG tanker traffic—and cumulative air pollution impacts for nearby communities. The Gulf Coast is also subject to sea level rise, subsidence, and extreme weather, such as hurricanes, tornadoes, and flooding, that can damage LNG terminals, creating safety concerns. Beyond the safety risks

---

[17] *Id.*

[18] FERC, *U.S. LNG Export Terminals – Existing, Approved not Yet Built, and Proposed* (May 28, 2025).

[19] *See id.*; *Community & Explore*, City of Starbase-Texas, https://cityofstarbase-texas.com/community-explore (noting that SpaceX launches can be viewed from the nearby cite of Port Isabel).

[20] *See, e.g.*, Carolos Anchondo, *New Gas Export Terminal Planned for Texas Coast*, E&E News (June 30, 2025), https://subscriber.politicopro.com/article/eenews/2025/06/30/new-gas-export-terminal-planned-for-texas-coast-00429556 (noting planned carbon capture at Coastal Bend LNG, Cameron LNG, and CP2 LNG).

caused by accidents or natural disasters, there are also the risks of intentional acts of terrorism targeting LNG facilities.

ii. Greenhouse gas emissions from LNG facilities contribute to climate change that harms human health and the environment.

PHMSA is required to consider environmental protection when issuing standards for LNG facilities.[21] Each stage of the LNG supply chain creates significant environmental impacts, including product leakage, end-use combustion, and the energy-intensive liquefaction process.

The urgency of the crisis resulting from climate change and the impact of human activities on global warming, are both clear. As the Intergovernmental Panel on Climate Change (IPCC) stated: "Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming… . Global surface temperature has increased faster since 1970 than in any other 50-year period over at least the last 2000 years (*high confidence*)."[22] Climate impacts are being felt across the U.S. and the globe, and while "[m]any changes due to past and future greenhouse gas emissions are irreversible for centuries to millennia,"[23] extreme weather events are projected to be "larger in frequency and intensity with every additional increment of global warming."[24] The U.S. Global Change Research Program (USGCRP)—a longstanding federal program between EPA, NASA, NOAA, the National Science Foundation,

---

[21] 49 U.S.C. § 60102(b)(1)(B)(ii) requires the agency to consider environmental protection when issuing standards aimed at "risks to life and property posed by pipeline transportation and pipeline facilities." *See* 49 U.S.C. § 60101(a)(19), (a)(21)(A)(i), defining "pipeline transportation" as "transporting gas" and defining "transporting gas" as "the gathering, transmission, or distribution of gas by pipeline, or the storage of gas, in interstate or foreign commerce." Furthermore, when prescribing any standard—including under 49 U.S.C. § 60103 specific to LNG facilities—PHMSA must consider "environmental information," 49 U.S.C. § 60102(b)(2)(A). And PHMSA shall "propose or issue a standard under this chapter only upon a reasoned determination that the benefits, including safety *and environmental benefits*, of the intended standard justify its costs." 49 U.S.C. § 60102(b)(5) (emphasis added).

[22] IPCC, AR6 Synthesis Report: Climate Change 2023, Summary for Policymakers (2023), at A.1-A.1.1.

[23] IPCC, Climate Change 2021: The Physical Science Basis, Summary for Policymakers (2021), at B.5 [hereinafter AR6 2021].

[24] *Id.* at 18.

and others—concluded in 2018 that "evidence of human-caused climate change is overwhelming and continues to strengthen," "the impacts of climate change are intensifying across the country," and "climate-related threats to Americans' physical, social, and economic well-being are rising."[25] In its Fifth National Assessment published in 2023, the USGCRP found that "[t]he evidence for warming across multiple aspects of the Earth system is incontrovertible," and "[h]uman activities are changing the climate."[26]

Climate change is already devastating American communities, and will continue to do so, through increases in the frequency and intensity of extreme heat, heavy precipitation, agricultural and ecological droughts, and an increase in major hurricanes in the North Atlantic and tropical cyclones.[27] For example, "[o]ngoing drought amplified the record-breaking Pacific Northwest heatwave of June 2021, which was made 2° to 4°F hotter by climate change," and which led to over 1,400 heat-related deaths and total damages exceeding $38.5 billion (in 2022 $).[28] The USGCRP has found that climate change worsens "long-standing inequities, contributing to persistent disparities in the resources needed to prepare for, respond to, and recover from climate impacts," such as finding that "[e]xtreme heat can lead to higher rates of illness and death in low-income neighborhoods, which are hotter on average."[29]

---

[25] A. Jay et al., *Fourth National Climate Assessment Vol. II, Impact, Risks, and Adaptation in the United States, Chapter 1: Overview*, U.S. GLOBAL CHANGE RESEARCH PROGRAM, at 36 (Reidmiller et al. eds.) (2018).
[26] USGCRP, *Fifth National Climate Assessment*, A.R. Crimmins et al., Eds. (2023), available at https://repository.library.noaa.gov/view/noaa/61592.
[27] *See* IPCC, AR6 2021 at 15, B.2; J.P. Kossin et al., *Global increase in major tropical cyclone exceedance probability over the past four decades*, 117 PROCEEDINGS NAT'L ACAD. SCI. 11975 (June 2, 2020), https://www.pnas.org/content/117/22/11975.
[28] USGCRP, *Fifth National Climate Assessment* at 1-18 through 1-19 (2023).
[29] *Id.* at 1-19.

Methane, the primary component of natural gas—including liquified natural gas[30]—is a potent greenhouse gas (GHG) that has approximately 83 times the global warming potential of carbon dioxide over a 20-year timeframe, and approximately 30 times the warming potential of carbon dioxide over a 100-year timeframe.[31] And when methane is burned, either at its end-use destination or when flared at any point along the supply chain, it releases climate-warming carbon dioxide emissions. To mitigate the harmful effects of climate change, it is imperative that humans reduce reliance on and combustion of fossil fuels, including natural gas, and that the oil and gas industry rapidly reduce methane emissions from the supply chain.[32]

Further, significant sources of GHG emissions at liquefaction facilities include emissions from the large amount of energy needed to run liquefaction equipment, as well as flaring, which occurs during plant startups and shutdowns.[33] A recent report by the Internation Energy Agency found that liquefaction accounts for 33% of the global LNG export industry's GHG emissions.[34]

Moreover, any leak of methane along the entire natural gas supply chain—extraction, processing, transportation via pipeline, liquefaction, transfer to ships for exported LNG, regassification—also contributes to climate change. In its May 2023 proposed rule regarding Gas Pipeline Leak Detection & Repair (which has yet to be finalized), PHMSA presented detailed estimates of methane emissions from LNG facilities, explaining that "methane regularly escapes

---

[30] *See* PHMSA, Proposed Rule: *Pipeline Safety: Gas Pipeline Leak Detection & Repair*, 88 Fed. Reg. 31,890, 31,899 (May 18, 2023) ("LNG means natural gas or synthetic gas having methane as its principal constituent, and which has been changed to a liquid").

[31] *See* IPCC, AR6 2021.

[32] *See* Ocko et al., *Acting rapidly to deploy readily available methane mitigation measures by sector can immediately slow global warming*, 16 ENV'T RSCH. LETTERS 054042 (2021), https://iopscience.iop.org/article/10.1088/1748-9326/abf9c8.

[33] International Energy Agency, *Assessing Emissions from LNG Supply and Abatement Options*, at 12–13, https://iea.blob.core.windows.net/assets/df9b1bed-4e16-4db8-bec5-a5fa117a9130/AssessingemissionsfromLNGsupplyandabatementoptions.pdf.

[34] *Id.* at 4, fig. 1.

from LNG facilities through compressor rod packing and valve leakage, incomplete combustion during flaring, and other various process venting sources."[35]

The warming climate, caused in part by fossil-based energy sources like LNG, also presents unique risks to LNG facilities themselves and creates a negative feedback loop of safety and environmental harms. GHG emissions from the LNG export industry contribute directly to climate change, which increases the risk of severe weather on the Gulf Coast, which in turn increases safety concerns for coastal LNG export terminals.

PHMSA must consider these environmental impacts and the potential for updated regulations to minimize climate pollution by reducing flaring and leaks, whether through increased leak inspections or monitoring, improved engineering standards, or more protective leak repair requirements.

B. Safety incidents at LNG facilities demonstrate the need for improved regulations.

PHMSA must also update its LNG safety regulations to incorporate lessons learned from safety incidents that have occurred at LNG facilities in the past decade. These incidents highlight existing gaps in PHMSA's safety regulations, including shortcomings related to a lack of communication with the public and with first responders.

One prominent example is the June 8, 2022 explosion at the Freeport LNG facility located in Quintana, Texas and related public safety failures.[36] There, thermal expansion within an LNG line with an isolated pressure relief valve caused a loss of containment and a boiling

[35] PHMSA, Proposed Rule: *Pipeline Safety: Gas Pipeline Leak Detection & Repair*, 88 Fed. Reg. 31890, 31905-906 (May 18, 2023).
[36] *Freeport LNG Incident – June 2022*, FERC, https://www.ferc.gov/industries-data/resources/project-directory/freeport-lng-incident-june-2022.

liquid expanding vapor explosion.[37] The flammable methane vapor that was released ignited, resulting in a vapor cloud explosion, a fireball, a secondary pool fire, and a release of vaporizing LNG from damaged piping.[38] The pool fire burned for 30 minutes and led to particulate-laden smoke in the area.[39] The explosion created a fireball 450 feet high and 350 feet wide,[40] shook nearby homes, and threw lifeguards at nearby Quintana Beach from their chairs.[41] No emergency sirens or evacuation orders were sounded, however, and residents nearby, who heard a sound like rolling thunder and were able to see smoke and flames rise above the plant, only learned the details of the incident when it was reported on local news.[42] The explosion resulted in the shutdown of the plant for eight months before it was allowed to partially return to service.[43]

While PHMSA found that violations of existing regulations contributed to the incident, its existing regulations were insufficient to inform the public of the facility's risks and ensure that the public was aware of the incident and response. Moreover, the incident can also inform the development of improved regulations in the future. PHMSA's investigation found that the direct cause of the incident was overpressurization of an 18-inch segment of pipe.[44] The overpressure protection for the segment had not been properly restored following annual testing several days earlier. As a result, LNG within the isolated segment of pipe warmed and expanded

---

[37] PHMSA/FERC/USCG, *Freeport LNG Incident and Regulatory Response,* at page 11, https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/2023-02/PHMSA-FERC-USCG%20Feb%2011%202023%20Freeport%20Information%20Session%20Presentation%20%28002%29_0.pdf.

[38] *Id.*

[39] PHMSA, Amended Notice of Probable Violation and Proposed Civil Penalty (Dec. 20, 2024), https://primis.phmsa.dot.gov/enforcement-documents/42024033NOPV/42024033NOPV_PCP%20(AMENDED)_12202024_(22-245663)_text.pdf.

[40] *Id.*

[41] Dylan Baddour & Delger Erdenesanaa, *Natural Gas Terminal Reopens After Massive Explosion*, Texas Observer (Feb. 23, 2023), https://www.texasobserver.org/freeport-lng-reopens-methane-explosion/.

[42] *Id.*

[43] Letter order granting Freeport LNG Development, L.P.'s 02/13/2022 request to commission, including cooldown, of Liquefaction Train 2 etc., Accession #: 20230221-3045, FERC Docket CP03-75 (Feb. 21, 2023), https://elibrary.ferc.gov/eLibrary/filelist?accession_number=20230221-3045.

[44] PHMSA Amended Notice of Probable Violation and Proposed Civil Penalty, *supra* note 39.

until it resulted in overpressurization and explosion, which caused a cascading series of multiple piping failures.[45] The explosion was ignited by contact between the flammable methane vapor and open and damaged electrical conduits and circuitry.[46] PHMSA issued a $1.5 million fine[47] for violations of its existing LNG safety regulations.[48] PHMSA should consider what amendments to its regulations could reduce the risk of similar incidents in the future.

Another incident occurred at the Sabine Pass LNG export terminal in 2018, a facility originally used for LNG imports. Workers discovered a release of LNG from a tank that caused cracks in the outer tank wall and pooling of LNG in the secondary containment area surrounding the tank.[49] The LNG, which is held in tanks in its cryogenic state at -260° F, cooled the outer tank wall to far below its design temperature of -25° F, causing cracks of up to six feet in length.[50] PHMSA investigators discovered a second tank that was also leaking methane gas vapors at fourteen points along the tank's base.[51] An estimated 245 barrels (~10,000 gallons) of LNG leaked from the cracks, releasing 825,000 cubic feet of gas into the atmosphere.[52] Two of the tanks at the facility remained closed for years following the incident, and PHMSA assessed a $2.2 million fine against the company.[53] One of the vulnerabilities exposed by this incident is that the tanks at Sabine Pass LNG, which was originally built as import/regassification terminal

---

[45] *Id.*

[46] *Id.*

[47] Mike Soraghan, *Texas LNG plant agrees to $1.5M fine for explosion,* E&E News (Feb. 6, 2025), https://www.eenews.net/articles/texas-lng-plant-agrees-to-1-5m-fine-for-explosion/.

[48] PHMSA Amended Notice of Probable Violation and Proposed Civil Penalty, *supra* note 39.

[49] PHMSA, Corrective Action Order, CPF No. 4-2018-3001H (Feb. 8, 2028), https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/docs/news/57286/420183001hcaosabine-pass02082018.pdf.

[50] *Id.*

[51] *Id.*

[52] PHMSA, Notice of Probable Violation Proposed Civil Penalty, CPF 4-2021-002-NPOV (July 1, 2021), https://primis.phmsa.dot.gov/enforcement-documents/42021002NOPV/42021002NOPV_PCP_07012021_(20-197852).pdf.

[53] Mike Soraghan, *Feds hit Cheniere with $2M fine over LNG leak*, EnergyWire (Aug. 10, 2021), https://subscriber.politicopro.com/article/eenews/2021/08/10/feds-hit-cheniere-with-2m-fine-in-lng-leak-279429.

and then converted for liquefaction/export, were only built to withstand temperatures as cold as -25° F,[54] while the walls of newer tanks at purpose-built export facilities are typically built to withstand temperatures as cold as -260° F.[55] Accordingly, PHMSA should consider what additional operational and maintenance regulations may be necessary to ensure safety at LNG import facilities converted to export facilities with tanks that do not comply with current regulations or industry standards for tank design.

Incidents have also occurred at peak shaving plants, notably the 2014 catastrophic failure and explosion at a peak shaving plant in Plymouth, Washington, which resulted in five injuries, including one hospitalization.[56] During routine annual liquefaction startup, an air-gas mixture in the piping auto-ignited.[57] The debris from the explosion damaged the outer and inner shell of a storage tank, various other components and buildings at the facility, and nearby railroad tracks.[58] The damaged tank allowed boil-off gas to escape, and communities within a two-mile radius were ordered to evacuate.[59] LNG was released for approximately 25 hours, totaling an estimated 234 barrels.[60]

These incidents confirm that PHMSA's existing regulations are insufficient and that the agency must increase monitoring and enforcement and make more protective its standards for operations, maintenance, inspections, emergency planning, and communication with the public and first responders.

---

[54] PHMSA, Corrective Action Order, CPF No. 4-2018-3001H, at 1, 2.

[55] Soraghan, *supra* note 53.

[56] Failure Investigation Report – Liquefied Natural Gas (LNG) Peak Shaving Plant, Plymouth, Washington (Apr. 28, 2016), https://www.phmsa.dot.gov/sites/phmsa.dot.gov/files/docs/FIR_and_APPENDICES_PHMSA_WUTC_Williams_Plymouth_2016_04_28_REDACTED.pdf.

[57] *Id.*

[58] *Id.*

[59] *Id.*

[60] *Id.*

C. PHMSA's regulatory revisions must account for and protect against additional operational risks—both studied risks and identified but not yet studied risks—that the ANPRM fails to address.

PHMSA has previously identified numerous knowledge gaps that exist with regard to LNG facility safety, including regarding the risk of vapor cloud explosions. The questions posed in PHMSA's ANPRM, however, do not address any of these identified knowledge gaps or the need for additional research, but are narrowly focused on costs to industry and benefits from updated industry standards incorporated within the regulations. PHMSA must consider the best available science that fills these previously-identified knowledge gaps in order to promulgate adequately protective LNG safety regulations. Additionally, to the extent that previously identified knowledge gaps still exist in the scientific and technical literature, PHMSA must conduct or commission new research and modeling in order to properly inform its update to the LNG safety rules.

PHMSA must consider the best available information to craft rules that mitigate the risk from vapor cloud explosions, which occur when a cloud of vapor ignites, especially in an area of congestion or confinement.[61] Vapor cloud explosions can damage nearby equipment that is engulfed in the explosion and lead to cascading failures.[62] Vapor cloud explosions can also generate projectiles, such as debris, pieces of piping, or larger equipment.[63] Methane, which is the primary component of natural gas and therefore is present at all LNG facilities, can create a deflagration vapor cloud explosion.[64] Other hydrocarbons, such as the refrigerants present at LNG export terminals, pose the risk for both deflagration and detonation explosions; these

---

[61] DNV & Thornton Tomasetti, *Assessing Cascading Effects from Flammable VCE: Technical Summary Report*, at 1 (prepared for PHMSA) (Dec. 20, 2022), https://primis.phmsa.dot.gov/matrix/FilGet.rdm?fil=17573&s=939FE54BE0A24FBF99B2E28E699A6EC3&c=1.

[62] *Id.*

[63] *Id*. at 6.

[64] *Id.*

detonation explosions may occur inside or outside areas of confinement.[65] A deflagration explosion is one in which flames move quickly, but at a speed slower than the speed of sound.[66] A detonation explosion is one that produces a "shock wave" that can initiate additional combustion.[67] Flammable fuels typically present at LNG facilities include methane, ethane, ethylene, propane, butane, pentane, and hexane.[68]

PHMSA's amended regulations must incorporate recommendations from the 2022 report prepared for PHMSA on vapor cloud explosions, which noted that export facilities often contain large areas of congestion around the liquefaction equipment, and that "[i]f congested regions are close together without adequate separation distances, they may combine and act as one large, congested region in a deflagration incident."[69] Some export facilities surveyed for that report contained over 1 million square feet of "congested" area that created a heightened risk of explosion.[70] The report concluded that, based on the findings from its detailed modeling, "guidance can be developed to comment on separation distances, deflagration versus detonation differences, volume congestion and confinement and the relation to cascade effects."[71] The report stated that, at facilities that have only methane/LNG onsite, "the VCE hazard is limited to deflagration and the key elements to limit cascade effects will be equipment spacing and levels of congestion and confinement."[72] At facilities with other flammable hydrocarbons onsite, the

---

[65] *Id*. at 7.
[66] *Id.* at 15.
[67] *Id*. at 16.
[68] *Id.* at 20.
[69] *Id.*
[70] *Id.*
[71] *Id*. at 114.
[72] *Id*.

focus for reducing potential detonation hazards would be "on siting of hydrocarbon inventories and understanding of the potential dispersion range of the flammable clouds."[73]

Similarly, the 2022 report recommends that hazards from projectiles that could lead to cascading effects could be mitigated by "advice on siting of equipment (such as cylinders) that may become projectiles."[74] The size of a vapor cloud created by a release or leak depends on a number of factors, including wind speed and ambient temperatures; larger clouds are created at lower wind speeds and higher ambient temperatures.[75] The chance of a vapor cloud from a given leak igniting could be hundreds of times greater in no-wind conditions because the vapor cloud would spread over such a greater area and therefore a greater number of potential ignition sources.[76]

Because LNG export terminals increasingly comprise large numbers of liquefaction trains located on the same site and are increasingly located in close proximity to other LNG terminals, PHMSA must consider what amendments to its rules are necessary to minimize the risks of vapor cloud explosions creating cascading failures. An increased number of liquefaction trains means more components that could potentially leak or fail and greater amounts of methane or refrigerants that could leak. It also means larger contiguous congested areas that could lead to deflagration, as well as a greater number of components that could be engulfed in an explosion and trigger a cascading failure. Regulations that may have been sufficient to ensure safety at terminals with one or two liquefaction trains are likely not sufficient at a facility of eight or nine

---

[73] *Id.*
[74] *Id.*
[75] *Id.* at 38; Graham Atkinson & Jonathan Hall, Health & Safety Laboratory, *Review of Vapor Cloud Explosion Incidents* at 7 (June 2016), https://primis-meetings.phmsa.dot.gov/archive/MH_15_80_HSL_Review_of_VCE_incidents_-_July_13th_2016.pdf.
[76] Atkinson & Hall, *supra* note 75, at 7.

liquefaction trains, and even less so when located less than a mile away from another LNG liquefaction and export facility.

PHMSA's updates to its LNG regulations should incorporate the knowledge gained from the numerous research projects PHMSA has sponsored on topics related to LNG. For example, two ongoing PHMSA-sponsored research projects are "Gap Analysis to Identify Opportunities to Inform Leak Detection and Repair (LDAR) Programs in LNG Facilities"[77] and "Liquefied Natural Gas (LNG) Knowledge Development, Consequences of Catastrophic Failure of LNG Storage Tanks."[78] Two recently-completed PHMSA-sponsored research projects are "Development of Methodologies for Managing Process Safety Risk Using Hazard and Operability (HAZOP) and Layers of Protection Analysis (LOPA) for LNG Facilities,"[79] and "LNG Knowledge Development, Develop Methodologies for Cryogenic and Fireproofing Requirements for LNG Facilities."[80] PHMSA also contracted with Sandia National Laboratories to develop a *Model Evaluation Protocol for Fire Models Involving Fuels at Liquefied Natural Gas Facilities.*[81] Other PHMSA-sponsored research projects on LNG safety topics include

---

[77] *Gap Analysis to Identify Opportunities to Inform Leak Detection and Repair (LDAR) Programs in LNG Facilities,* Project No. 1042, PHMSA Research & Development Program, https://primis.phmsa.dot.gov/matrix/PrjHome.rdm?prj=1042&s=03C483D7F6A44EDE94DC2D89DECBE43B&c=1.

[78] *Liquefied Natural Gas (LNG) Knowledge Development, Consequences of Catastrophic Failure of LNG Storage Tanks*, Project No. 1041, PHMSA Research & Development Program, https://primis.phmsa.dot.gov/matrix/PrjHome.rdm?prj=1041&s=03C483D7F6A44EDE94DC2D89DECBE43B&c=1.

[79] *Development of Methodologies for Managing Process Safety Risk Using Hazard and Operability (HAZOP) and Layers of Protection Analysis (LOPA) for LNG Facilities,* Project # 1007, PHMSA Research & Development Program, https://primis.phmsa.dot.gov/matrix/PrjHome.rdm?prj=1007&s=03C483D7F6A44EDE94DC2D89DECBE43B&c=1.

[80] *LNG Knowledge Development, Develop Methodologies for Cryogenic and Fireproofing Requirements for LNG Facilities,* Project # 1010, PHMSA Research & Development Program, https://primis.phmsa.dot.gov/matrix/PrjHome.rdm?prj=1010&s=03C483D7F6A44EDE94DC2D89DECBE43B&c=1.

[81] Anay Luketa, Sandia National Laboratories, SAND2022-6812, *Model Evaluation Protocol for Fire Models Involving Fuels at Liquefied Natural Gas Facilities* (May 2022), https://www.osti.gov/biblio/1869376.

Comparison of Exclusion Zone Calculations and Vapor Dispersion Modeling Tools.[82] Significant resources and taxpayer dollars have already been expended to identify and, in some cases, fill knowledge gaps in the literature, and it would be arbitrary for PHMSA to fail to consider all of these studies in updating its LNG regulations.

PHMSA should consider constructive recommendations that were offered in multiple presentations[83] at the PHMSA Public Workshop on Liquefied Natural Gas (LNG) Regulations[84] held in 2016. Presenters explained the importance of accounting for all hazards posed by an LNG facility, including flammable vapor cloud dispersion, fire radiation, explosion, and toxic vapor clouds, among others.[85] They also recommended consideration of the size and layout of individual facilities, the failure rates of various components within an LNG facility, and the local weather conditions.[86]

PHMSA must also ensure that all risk modeling employed is subject to peer review and public vetting. One scientist submitted a comment on the 2016 workshop pointing out gaps related to modeling of vapor cloud explosion risk, which relies on models that are typically proprietary and thus not subject to peer review or public vetting. [87] He provided one example of a

---

[82] *Comparison of Exclusion Zone Calculations and Vapor Dispersion Modeling Tools*, Project No. 640, PHMSA Research & Development Program, https://primis.phmsa.dot.gov/matrix/PrjHome.rdm?prj=640.

[83] Jeff Marx, Quest Consultants Inc., *Quantitative Risk Analysis (QRA) for LNG Facility Siting*, https://primis-meetings.phmsa.dot.gov/archive/19th_May_-_7_-QRA_for_Siting_Marx.pdf (File No. 7833, PHMSA Meeting Archive); Cheryl Stahl, *Risk Based Siting for Small Scale LNG: A Comparison of Alternatives,* (May 19, 2016), https://primis-meetings.phmsa.dot.gov/archive/19th_May_-_8_Risk_Based_Siting_DNV_GL_Stahl.pdf (File No. 770, PHMSA Meeting Archive); Sara Gosman, *LNG Facility Siting, Risk, and the Public* (May 18, 2016), https://primis-meetings.phmsa.dot.gov/archive/18th_May_-_4_Facility_Siting_Risk_and_the_Public_Gosman.pdf (File No. 763, PHMSA Meeting Archive).

[84] *PHMSA Public Workshop on Liquefied Natural Gas (LNG) Regulations,* PHMSA Public Meetings, https://primis-meetings.phmsa.dot.gov/archive/MtgHome.mtg@mtg=111.html (Meeting Archive).

[85] Marx, *supra* note 83 at 9.

[86] *Id.*

[87] Comment of Jerry Havens on *Pipeline Safety: Public Workshop on Liquefied Natural Gas Regulations,* 81 Fed. Reg. 24,689 (Apr 26, 2016), Comment ID PHMSA-2016-0005-0006 at 1–2, https://www.regulations.gov/comment/PHMSA-2016-0005-0006.

sub-model, known as SOURCE5, being discredited after independent review, but noted that newer models are more expensive and therefore not as likely to be subject to the same level of public criticism that led PHMSA to commission an independent review of SOURCE5.[88] Vapor cloud explosion models also might not be accurately accounting for the gas-impervious vapor fences LNG export terminals have recently started using.[89] These fences are intended to contain any potential vapor cloud within the facility property line, but fences could also have the effect of increasing the concentration of flammable vapor within the fence-confined area, therefore increasing the potential for an explosion.[90]

PHMSA must also ensure that it addresses previously-identified knowledge gaps as part of and as the basis for comprehensive updates to its standards for LNG facilities. For example, the working group report-outs from the 2022 LNG Research and Development Public Meeting and Forum[91] held by PHMSA identified numerous research gaps:

- Research gaps related to LNG facility design and construction[92]:
    - "Develop criteria for cryogenic and fire protection requirements based on hazard modeling study."
    - "Modern plant designs usually include a mix of manual, remote, and automatic valving, but each design is different. Research to determine the appropriate locations for remote control valves vs manual valves in designing a plant, to better protect personnel safety and minimize volumes of a release."
    - "Effective use of remote impoundment of methods, pipe in pipe, spacing/storage to mitigate risks of chain failure events and fire."

---

88 *Id.* at 2–3.
89 *Id.* at 3.
90 *Id.* at 3–5.
91 *Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum,* PHMSA Public Meetings, https://primis-meetings.phmsa.dot.gov/archive/MtgHome.mtg@mtg=162.html; *see also Pipeline Safety: Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum,* 87 Fed. Reg. 64,539 (Oct. 25, 2022).
92 *Working Group 1 Report-Out,* Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum, https://primis-meetings.phmsa.dot.gov/archive/Working_Group_1_Report-Out.pdf.

- “Technologies for reducing the hydrogen concentrations (to an acceptable level) in the feed gas to existing LNG facilities.”
- “Study spill collection of LNG piping near waterways. 1.) The effectiveness of different collection systems prevention LNG from entering the waterway for different hole size and operation pressures and 2.) the overpressure cause by rapid phase transition of LNG which has entered the waterway.”
- “Effectiveness of different isolation philosophies, and when to use them (i.e., double block and bleed, what piping pressure class to use DBB? When is removable spool preferred vs a blind?).”
- “[H]ow would PHMSA adapt new innovative technologies for geotechnical investigation for the LNG facility site?”
- “Guidance on quantitative and semi-quantitative criteria to perform risk reduction using HAZOP and LOPA. Guidance could be in form of FAQ or other means.”
- “PHMSA development and research on Piping stress analysis regarding vibration impacts on small piping system under 2 inches.”
- “Minimum amount of Hydrogen (H2) that can be exposed to plant materials that would be small enough too not affect materials by HSCC. Of particular concern is 9% Ni Steel.”
- “Overall impacts of hydrogen in pipelines to the entire LNG facility. Do codes and standards used for LNG facilities need to be updated for hydrogen?”
- “Research and development regarding the design of Seismic Isolator for the LNG storage tank in the high seismic region.”
- “Impact of hydrogen concentration, pressure and temperature on pipeline integrity.”
- “Lab or in-field experiment to validate the thermodynamic methods, commercially available software (process and computational fluid dynamics (CFD)) outputs H2 has on the liquefaction process, LNG storage tank, vapor space, liquid space, and top/bottom fill.”
- “Develop guidance for conducting Preliminary Hazard Analysis (PHA) of LNG facilities (small scale and large scale) and require LNG facilities to conduct PHA prior to construction and PHA revalidations at least every 5 years.”
- “Pipeline structure safety design and flaw detection under extreme loading conditions, which considers multiaxial and complex stress loadings, hydrogen embrittlement, and material aging.”
- “Qualify and quantify environmental injustice in siting and design choices and what should be done to enact justice.”

- “Review relief system design basis and calculations and determine how hydrogen enriched LNG will affect relief valve overpressure scenarios, sizing calculations, etc.”
- “Review incident history of hydrogen induced embrittlement at refineries and develop best practices that should be applied in the design and construction of LNG facilities (i.e., materials of construction) and ways to detect/monitor for hydrogen embrittlement scenarios.”
- “Liquefaction – Evaluation of applicable refrigerants and changes in the environmental impact as available blends change.”
- “Hurricane/severe weather design research question focusing on whether environmental parameter thresholds underlying existing standards will remain valid for decades in light of climate change and more severe extreme storms.”.
- “Design considerations for safety to include: explosion risks of LNG tanks and refrigerant tanks and then what happens when there is an explosion (Emergency Response Plan).”
- “Design considerations for shutdown and malfunction events: how can there be zero air emissions after a storm for example.”
- “Cumulative effects: pollution (air, water, noise) including fugitive emissions of operation, construction, and malfunction of facilities; effects on workers, surrounding communities and public health.”
- “Pipeline monitoring for hydrogen carrier pipelines – leak detection including odorization.”
- “If H2 is removed from LNG feedstock, how low can the levels be brought down to "economically?”
- “Evaluating the minimum number of fixed points spaced around the circumference and within interior a tank needs such that it does not overstress, cause buckling, or exceed other API limits.”
- “Specific siting guidelines for Gravity Based Structures (GBS) for LNG storage, including storage tanks mounted on top of GBS and GBS where the LNG is stored inside the concrete caisson. Facility Siting should consider the potential role of GBS in enhancing a creatinine retaining walls.”
- “Development of computational tools or models based on fracture mechanics or material corrosion will help to evaluate and monitor the vessel safety.”
- “It is likely that open expansion LNG facilities would be most affected. These facilities would need to be redesigned for each HENG H2 concentration – is there any recourse for such facilities other than to switch to a non-open expansion system?”

    - "Design considerations for decommissioning all facilities (including a cost-benefit analysis to include the lifespan of the terminal and pipelines); LCA of Pipeline Replacement."
- Research gaps related to LNG facility siting[93]:
    - "Acceptable methods of modeling shrouding or obstructions for liquid leaks."
    - "Exploring barriers and materials that can be used to mitigate thermal radiation extending off-property."
    - "Impact of cascading explosions within LNG site and understanding relevant threshold limits."
    - "Establishing risk tolerability criteria."
    - "Validation data for large LNG fires (size of storage tank) not performed over water."
- Research gaps related to LNG facility fire protection system design[94]:
    - "Conduct pool fire tests at scales typical to LNG impoundment installations to provide data sets to support model validation and quantify the performance of pool fire mitigation measures."
    - "Water spray and/or curtain design validation to ensure effectiveness and suggested locations for fire and vapor dispersion mitigation based on two scenarios: 1) to ensure vapor dispersion mitigation; 2) blocking pool fire thermal radiation."
    - "Hydrogen blending impacts on hazard detection and other hazard mitigation designs/installations and failure modes and likelihood."
    - "Experiments to validate the integrity and effectiveness of passive pool and jet fire mitigation systems; guidance on fire scenarios to be included in passive mitigation design and guidance on failure criteria for pressure vessels and steel."
    - "Continuous monitoring of LNG infrastructure using long open-path standoff gas detection to provide a low-cost and flexible solution. Looking at gas leaks in facilities. Looking at temperature cracks, gas detection for safeguarding facilities; open path detectors."
    - "Develop guidance / scenario framework for fire protection evaluation so operator can have a better idea on how to approach fire protection system

[93] *Working Group #2: Liquefied Natural Gas (LNG) Facility Siting Summary Report-Out*, Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum, https://primis-meetings.phmsa.dot.gov/archive/Working_Group_2_Report-Out.pdf

[94] *Working Group #3: Liquefied Natural Gas (LNG) Facility Fire Protection System Design Summary Report-Out*, Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum, https://primis-meetings.phmsa.dot.gov/archive/Working_Group_3_Report-Out.pdf

design; Subtopic: Evaluation of necessary use of water protection systems (small scale LNG facility).”
    - “Low-cost gas leak imager for surveying LNG tanks and LNG process piping.”
    - “Radiant heat time based thresholds where typical steel fails.”
    - “Experimental data or validation for basis of design and distances used in electrical area classification consistent with basis in other standards such as NFPA 497 and American Petroleum Institute 500.”
    - “Conduct pool fire tests at scales typical to LNG impoundment installations to provide data sets to support model validation and quantify the performance of pool fire mitigation measures.”
- Research gaps related to LNG facility operation and maintenance[95]:
    - “Available technologies to enhance inspection of in-service legacy LNG storage tanks (Part 193.2623 Inspecting LNG storage tanks).”
    - “Potential impacts of hydrogen-enriched natural gas feedstock on existing LNG storage tank systems (e.g., determine the potential for stratification/rollover, whether the H2 stays in the LNG to some degree, and whether the H2 becomes concentrated in various vapor spaces, such as the dome or annular space, within various storage tank styles).”
    - “Would an OSHA PSM performance-based approach for establishing Recognized and Generally Accepted Good Engineering Practice (RAGAGEP) provide a higher level of protection in concert with risk-based regulations, as addressed in PIPES 2020 Section 110 for LNG facilities?”
    - “Update weld design, processes, testing, and assessment procedures for LNG structural pipes and storage containers.”
    - “Develop standardized stratification thresholds based on operating mode and across facilities (monitoring and alarm levels for temperature and density).”
    - “Recommended allowable concentrations of trace components in renewable natural gas (e.g., siloxanes--perhaps Br, Cl, HBr, HCl or others) in feed gas to LNG liquefaction systems.”
    - “Effectiveness of relief valves (hard seated/soft seated) to seal below set lift pressure for gas streams that are potentially enriched (1-5%) with hydrogen.”

[95] *Working Group #4: Liquefied Natural Gas (LNG) Facility Operation and Maintenance Summary Report-Out*, Liquefied Natural Gas (LNG) Research and Development (R&D) Public Meeting and Forum , https://primis-meetings.phmsa.dot.gov/archive/Working_Group_4_Report-Out.pdf.

- “Improved methods of flaring off excess vaporized natural gas through flare-columns (e.g., replacing the sweep gas with cleaner burning gas such as hydrogen, capturing the excess gas for redistribution in the system, or utilizing carbon capture with the flare column).
- “Expected life of an LNG tank that never cycles to empty.”

In summary, there are important data sources and information on which PHMSA can rely to identify the gaps in its current LNG safety regulations, including PHMSA-funded research, other research, and investigation reports into incidents at LNG facilities. In some cases, these sources identify operational practices or design criteria that can improve safety and that PHMSA should incorporate into its regulations. In the event PHMSA does not adopt those recommendations, it must explain the basis for rejecting such safety measures. In other cases, these sources identify the need for further research, which PHMSA should provide for as well. PHMSA cannot move forward on a comprehensive update of its LNG safety rules without considering the information in all these sources and the need for future research identified within them.

III. PHMSA must ensure adequate opportunity for public engagement on any updated LNG safety rules.

While most of the questions posed in PHMSA’s ANPRM seem designed to elicit information from LNG facility operators, it is crucially important that PHMSA solicit public input and provide adequate opportunity for public engagement during the development of its updated LNG safety rules. PHMSA cannot move forward with a proposed rule without ensuring that it receives input from the affected members of the public who face the greatest risk to their lives and property from operation of LNG facilities, including local community members, first responders, and workers at the facilities. PHMSA must also ensure it receives input from members of the public concerned about the environmental harms posed by LNG facilities.

PHMSA should hold public hearings as part of the comment process on this ANPRM and on any notices of proposed rulemaking (NPRM) that it may issue on its LNG safety rules. While PHMSA's ANPRM makes only a vague suggestion that the agency will "hold a public meeting in the near future to supplement or to clarify the materials received in response to this ANPRM," 90 Fed. Reg. 18,951, PHMSA should hold at least one public hearing at which individuals can give oral comments as part of the comment period on this ANPRM and on any NPRM it issues on its LNG rules. PHMSA should ensure that at least one such meeting is located within the Gulf Coast communities that face the greatest concentration of currently-operational, under-construction, and planned large-scale LNG export facilities.

IV. Responses to Specific Topics Under Consideration in the ANPRM.

A. Response to Topic C.3: PHMSA should clarify that reporting of all safety-related conditions is required.

Because PHMSA's statutory mandate to require reporting of safety-related conditions does not contain any exception for LNG facilities that are not "in service," PHMSA should make whatever amendments to its regulations are necessary to clarify that *all* safety-related conditions must be reported. PHMSA is required to "prescribe regulations requiring each operator of a pipeline facility (except a master meter) to submit to the Secretary a written report on any … (A) condition that is a hazard to life, property, or the environment; and (B) safety-related condition that causes or has caused a significant change or restriction in the operation of a pipeline facility." 49 U.S.C. § 60102(h)(1).

While the regulation currently does not contain any exceptions or carve-outs from the requirement that "[i]ncidents, safety-related conditions, and annual pipeline summary data for LNG plants or facilities must be reported," 49 C.F.R. § 193.2011, PHMSA asks about the costs and benefits that might "arise from amending § 193.2011 to clarify that safety-related condition reporting would be required not only for 'in-service' LNG facilities, but also safety-related conditions that occur during commissioning and initial start-up?" 90 Fed. Reg. at 18,951. To the extent that the regulation's coverage of all safety-related conditions, without exception, is not already clear, PHMSA should modify the regulation to clarify that point. PHMSA should also clarify that safety-related conditions that occur at any time, including during times other than in-service, commissioning, or initial startup—such as during de-commissioning—must be reported.

B. Response to Topic D.2: Adopting the 2023 version of NFPA 59A is a reasonable interim step while PHMSA works toward complying with the statutory mandates to update the LNG facilities rules.

While adopting the 2023 version of NFPA 59A is a reasonable interim step for PHMSA to take while it works toward a more comprehensive update to its LNG facilities rules, as explained above, *see supra* Section I, doing so would not be sufficient to fulfill either of PHMSA's two outstanding mandates to update its LNG facilities rules. Additionally, as explained above, *see supra* Section I, PHMSA would need to collect significantly more safety and environmental information, in addition to information on the topics listed in the ANPRM, to inform a comprehensive rulemaking that fulfills its statutory mandates. While PHMSA may be able to quickly adopt the updated version of NFPA 59A in order to provide safety benefits to the public and reduce redundancy to industry in the near term, PHMSA should take the time necessary to ensure that its comprehensive LNG safety rulemaking considers all the relevant safety, environmental, and engineering information to determine how to minimize safety risks and protect the environment to the greatest extent possible.

C. Response to Topic H: The executive orders directing agencies to reduce regulatory burdens do not change PHMSA's statutory mandates to regulate for safety and environmental protection and to specifically update its LNG regulations.

The cited executive orders do not provide a justification—let alone a requirement—for PHMSA to act contrary to its statutory mandates to issue updated LNG safety rules. As explained above, *see supra* Section I, PHMSA has a statutory obligation to issue minimum safety standards for LNG facilities and has been specifically directed by Congress in the PIPES Act of 2016 and the PIPES Act of 2020 to update its LNG safety rules in specified ways. Under Topic H, PHMSA asks a number of questions related to executive orders that it asserts "direct[] PHMSA and other agencies to reduce regulatory burdens on the U.S. energy industry," 90 Fed. Reg. 18,953, including Exec. Order No. 14192, *Unleashing Prosperity Through Deregulation*; Exec. Order No. 14154, *Unleashing American Energy*; and Exec. Order No. 14156, *Declaring a National Energy Emergency*. 90 Fed. Reg. 18,950. Because agencies are created by statute and their powers and duties are prescribed in statute by Congress, *Nat'l Fed'n of Indep. Bus. v. OSHA*, 595 U.S. 109, 117 (2022), *Marin Audubon Soc'y v. FAA*, 121 F.4th 902, 912 (D.C. Cir. 2024) (quoting *Fed. Election Comm'n v. Cruz*, 596 U.S. 289, 301 (2022)), an executive order can neither alter an agency's statutory authority nor relieve it of its statutory duties.

The President's authority to act, including authority to issue executive orders, "'must stem either from an act of Congress or from the Constitution itself.'" *Minnesota v. Mille Lacs Band of Chippewa Indians*, 526 U.S. 172, 188–89 (1999); *Youngstown Sheet & Tube Co. v. Sawyer*, 343 U.S. 579, 585 (1952). "When 'the President takes measures incompatible with the expressed or implied will of Congress ... he can rely only upon his own constitutional powers minus any constitutional powers of Congress over the matter.'" *Zivotofsky ex rel. Zivotofsky v. Kerry*, 576 U.S. 1, 10 (2015) (quoting *Youngstown,* 343 U.S. at 635–38 (Jackson, J., concurring)). "There is no provision in the Constitution that authorizes the President to enact, to amend, or to repeal statutes." *Clinton v. City of New York*, 524 U.S. 417, 438 (1998). Article II of the Constitution, to the contrary, directs the President to "take Care that the Laws be faithfully executed." U.S. CONST. art. II, § 3. Courts have found a violation of the separation of powers where a president issues an executive order that "does not direct that a Congressional policy be executed in a manner prescribed by Congress—it directs that a presidential policy be executed in a manner prescribed by the President." *Youngstown*, 343 U.S. at 588. Courts have also found that when agencies take action based on the *President's* priorities rather than *Congress's* "stated statutory factors," the "Executive has acted contrary to law and in violation of the APA." *New York v. Trump*, 764 F. Supp. 3d 46, 50 (D.R.I. 2025).

Therefore, whatever recent executive orders may say, PHMSA is statutorily obligated to regulate pipeline and LNG facilities with consideration for safety and environmental protection, 49 U.S.C. § 60102(a), (b), and to update its safety regulations for small scale and large scale LNG facilities pursuant to the PIPES Acts of 2016 and 2020. The executive orders likewise do not excuse PHMSA from complying with the requirements of the Administrative Procedure Act, including the requirement to provide a "rational connection between the facts found and the choice made," *Motor Vehicle Mfrs. Ass'n of U.S., Inc. v. State Farm Mut. Auto. Ins. Co.*, 463 U.S. 29, 52 (1983) (quoting *Burlington Truck Lines, Inc. v. United States,* 371 U.S. 156, 168 (1962)), and a justification for changing position. *Encino Motorcars, LLC v. Navarro*, 579 U.S. 211, 221 (2016).

Additional responses to the specific subtopics listed under Topic H are provided below.

Topic H.1 asks whether there are "current aspects of PHMSA regulations that are particularly burdensome on the ability of industry to operate LNG facilities that PHMSA should consider amending or rescinding." 90 Fed. Reg. at 18,953. Burden alone, however, cannot justify PHMSA's repeal of a safety rule. PHMSA can issue a rulemaking—including changes to its existing standards—"only upon a reasoned determination that the benefits, including safety and environmental benefits, of the intended standard justify its costs," 49 U.S.C. § 60102(b)(5), and thus PHMSA must consider both the costs a regulation imposes on industry and the safety and environmental benefits it confers on the public. PHMSA has clear statutory directives to set standards for pipelines and related facilities that ensure public safety and environmental protection, and Congress specifically directed PHMSA to ensure that its new regulations for large scale LNG facilities are at least as protective as current regulations, PIPES Act of 2020 § 110(b)(2). It would be inappropriate for PHMSA to remove regulations except upon a demonstration that the regulation is producing a burden on industry without producing any safety or environmental benefit, including: benefits to those living, working, or recreating near LNG facilities; benefits to individuals working at LNG facilities; safety during normal facility operation; safety during malfunction, fire, explosion or other incidents; transparency to the surrounding communities and local emergency responders; benefits to the environment from reduced emission of pollutants; and benefits to PHMSA from collecting information from facilities that can inform PHMSA's ongoing regulation of LNG facilities.

Topic H.2 asks whether there are "areas that PHMSA could add regulatory text that would reduce costs on operators without reducing safety outcomes." 90 Fed. Reg. at 18,953. As explained above, it would be potentially consistent with PHMSA's statutory directives to repeal or amend a regulation *if* PHMSA can conclusively demonstrate that doing so would not reduce safety or environmental benefits, including: benefits to those living, working, or recreating near LNG facilities; benefits to individuals working at LNG facilities; safety during normal facility operation; safety during malfunction, fire, explosion or other incidents; transparency to the surrounding communities and local emergency responders; benefits to the environment from reduced emission of pollutants; and benefits to PHMSA from collecting information from facilities that can inform PHMSA's ongoing regulation of LNG facilities.

Topic H.3. asks "[w]hich aspects of current PHMSA regulations do operators find outdated or unnecessary because industry practice or existing industry guidelines are an equivalent or better standard both for cost savings and safety outcomes?" 90 Fed. Reg. at 18,953. If any industry practices are leading to better safety or environmental outcomes than what is achieved by PHMSA standards, then PHMSA should consider updating its regulations to incorporate those best practices and ensure that, at a minimum, all industry members are following them. It would be inappropriate, and an abdication of PHMSA's statutory duty to establish minimum safety and environmental protection standards, 49 U.S.C. § 60102(a), (b); *Id.* § 60103(a), (b), (d), for PHMSA to merely remove regulations and leave it to the industry to voluntarily follow guidelines that are not legally enforceable. Codifying industry best practices provides important benefits because it prevents lagging operators from falling short of superior outcomes that leading industry actors may be achieving.

PHMSA is, of course, by no means limited to codifying only those safety practices that either are already common in the industry or are being used by a few industry leaders. PHMSA may adopt standards for siting, design, operations, and maintenance that are cost-benefit justified, 49 U.S.C. § 60102(b)(5), regardless of whether any currently-operational LNG facilities in the U.S. are already following those practices. Because it can take years for an LNG facility to be constructed and the facility may operate for decades, it is especially important for PHMSA to update its rules to include the most protective construction and design standards that are feasible and cost-effective for new facilities, even if facilities constructed years ago were not built to those standards.

Topic H.4 asks "[h]ow can PHMSA best design a rulemaking to update its part 193 regulations for LNG facilities to be deregulatory and lead to cost savings for the industry?" 90 Fed. Reg. at 18,953. As explained above, recent executive orders cannot override PHMSA's statutory duties and replace them with a requirement to take actions that are "deregulatory and lead to cost savings for the industry." Any LNG facility rule PHMSA issues must be consistent with its statutory mandate of setting standards "designed to meet the need for…safety …and…protecting the environment," 49 U.S.C. § 60102(b), and the mandates from Congress to update its LNG facilities regulations in the ways that Congress specified in the PIPES Act of 2016 and the PIPES Act of 2020. None of these mandates can be altered via executive order. Any rule PHMSA issues must also comply with the Administrative Procedure Act's requirement to explain a rational connection between the facts found and the choice made, including, if PHMSA changes position from a prior rulemaking, an explanation for the change in position. *State Farm*, 463 U.S. at 52; *Encino Motorcars*, 579 U.S. at 221. Additionally, any rulemaking PHMSA adopts is required by statute to comply with cost-benefit analysis requirements, which require consideration of environmental and safety benefits—and, therefore, environmental or safety harms. 49 U.S.C. § 60102(b)(5); *Interstate Nat. Gas Ass'n of Am. v. Pipeline & Hazardous Materials Safety Admin.*, 114 F.4th 744, 749–55 (D.C. Cir. 2024) (vacating "four standards for which PHMSA failed to make 'a reasoned determination that the benefits ... justify [the] costs.' 49 U.S.C. § 60102(b)(5)"). This means that if PHMSA wishes to issue a rule which removes current regulatory requirements, PHMSA must demonstrate that the benefit of removing the requirement outweighs any costs of doing so, including costs in the form of foregone safety or environmental benefits. It also means any "burden" PHMSA imposes on industry is required to be justified by the corresponding safety and environmental benefits to the public.

V. NEPA Scoping Comments

We provide these scoping comments to inform PHMSA's environmental analysis of its planned amendments to the LNG facilities rule under the National Environmental Policy Act (NEPA), 42 U.S.C. § 4321 *et seq.* PHMSA announced its intent to prepare an environmental impact statement (EIS) evaluating its planned rulemaking, including the environmental effects of incorporating by reference the 2023 edition of NFPA 59A, implementing the Congressional mandate from Section 27 of the PIPES Act of 2016, and implementing the Congressional mandate from the PIPES Act of 2020. *Pipeline Safety: Notice of Intent to Prepare an Environmental Impact Statement for the Amendments to Liquefied Natural Gas Facilities Rulemaking*, 90 Fed. Reg. 24,088 (June 6, 2025). PHMSA must complete an EIS, not merely an environmental assessment (EA). In considering alternatives to evaluate, PHMSA should be guided by the purpose and need to meet its outstanding Congressional mandates and to account for the technological advances, lessons learned, and changes in the industry since its LNG safety rules were last amended. Finally, PHMSA must consider the general categories of environmental impacts listed in its notice, as well as, at a minimum, the additional, more specific impacts noted below.

PHMSA must prepare an EIS on its rulemaking. Despite its announced intent to prepare an EIS, PHMSA notes that it will release a draft EIS "[i]f necessary" or "[i]f prepared," concurrently with its release of an NPRM. 90 Fed. Reg. at 24,088. This language seems to indicate that PHMSA is considering the possibility of preparing not an EIS, but an EA, instead. An EIS is required because amending PHMSA's LNG safety regulations would have a reasonably foreseeable significant effect on the human and natural environment. 42 U.S.C.

§ 4336(b)(1). The reasonably foreseeable, direct impacts of incidents such as pool fires and vapor cloud explosions are undoubtedly significant. Because the anticipated amendments to PHMSA's rules would, if properly designed, significantly reduce the likelihood of such incidents and/or mitigate their effects, PHMSA's rules would therefore have a significant impact. Additionally, amendments to PHMSA's requirements regarding various operations and maintenance practices may impact the frequency, duration, or character of flaring, use of security lights, or non-explosive leaks of methane or refrigerants, which would have a significant impact on the human and natural environment.

We agree that the purpose and need for PHMSA's amendments to its LNG safety rulemaking include the need to "address the significant changes that have occurred in the LNG industry in recent years" and to fulfill PHMSA's Congressional mandates by updating the regulations to "account for current technologies, best practices, and lessons learned, and to incorporate recent editions of consensus industry standards by reference." 90 Fed. Reg. at 24,089. While PHMSA states that its regulatory updates would also "advance[] President Trump's goal of unleashing American energy and achieving economic prosperity by alleviating unnecessary regulatory burdens," 90 Fed. Reg. at 24,089, as explained above, executive orders do not have the legal authority to override or contradict an agency's statutory mandate. *See supra* Section IV.C (Response to Topic H). Nor do the executive orders alter the facts on the ground that PHMSA acknowledges necessitate updates to its regulations. Therefore, PHMSA's purpose and need in completing this rulemaking and corresponding NEPA review should be defined to include meeting its Congressional mandates and updating its LNG rules to be protective of human safety and the environment, taking into account the changes in the industry, lessons learned, and updated research. Additionally, to the extent that PHMSA's amendments to its LNG facility safety rules include "deregulatory" actions, PHMSA must fully evaluate the environmental impacts of repealing any existing regulations, as compared to the current regulatory (no action) baseline.

The general categories of impacts that PHMSA anticipates evaluating in its EIS, 90 Fed. Reg. at 24,089, are all reasonably foreseeable impacts, consideration of which is necessitated by the statutory commands that PHMSA can issue a rulemaking "only upon a reasoned determination that the benefits, including safety and environmental benefits, of the intended standard justify its costs," 49 U.S.C. § 60102(b)(5), and that PHMSA must consider relevant safety and environmental information, *Id.* § 60102(b)(2). At a minimum, PHMSA must consider the following non-exhaustive list of more specific categories of impacts for each of its listed categories:

- Effects on populations located near LNG facilities:
  - Impacts of noise and light pollution from flaring of methane.
  - Impacts of light pollution from required security lighting.
  - Effects on local populations from all of the below categories of impacts.
- Impacts related to public safety and human health in the event of accidental release of hazardous substances from LNG facilities:
  - Impacts from releases resulting in pool fires and vapor cloud explosions, such as injuries to hearing from the loud noise of explosions, inhalation of particulate matter emitted by fires and explosions, burns, injuries from debris ejected from explosions, and secondary injuries (*e.g.*, falls, car accidents) that may occur as people hastily evacuate the area of a release. This should include impacts to workers at LNG facilities, nearby residents, and first responders.
  - Impacts related to releases of methane that do not ignite or combust.
  - Impacts related to releases of refrigerants or other chemicals present at LNG liquefaction facilities.
  - Cumulative safety impacts to communities located in close proximity to numerous LNG facilities and other industrial infrastructure.
- Effects on human health from factors such as emissions of air pollutants:
  - Impacts from emissions of methane and other greenhouse gases that contribute to climate change.
  - Impacts from emissions of particulate matter, VOCs, and other pollution.

    - Cumulative air pollution impacts to communities located in close proximity to numerous LNG facilities and other industrial infrastructure.
- Effects on socioeconomic factors such as economy, employment, transportation (e.g., road, and marine):
    - Impacts from increasing the number of required inspections, leading to increased demand for inspectors.
    - Impacts to nearby property values from increasing safety and decreasing pollution.
    - Impacts from increased safety leading to increased customer traffic to businesses located close to LNG facilities, or increased ability of fishermen to fish in areas located close to LNG facilities.
- Effects on historic and cultural resources:
    - Impacts that may affect people's ability to access historic cemeteries, sacred indigenous sites, and religious sites.
- Effects on land use:
    - Impacts to property values of surrounding residential or commercial property due to concerns about safety, air pollution, noise pollution, light pollution, and water pollution.
    - Impacts to people's ability to safely access recreation areas on land and water, including campgrounds, hunting areas, hiking or biking trails, wildlife viewing areas, sports facilities, beaches, lakes, rivers, and near-shore ocean waters.
- Effects on natural resources:
    - Impacts from emissions of methane and other greenhouse gases that contribute to climate change, including cumulative impacts.
    - Impacts from emissions of particulate matter and other pollution that affects waterways, plant growth, and other natural resources, including cumulative impacts to environments impacted by numerous LNG facilities.
    - Impacts to wetlands, streams, groundwater, and oceans, including cumulative impacts to waterways impacted by numerous LNG facilities.
    - Impacts to wildlife, including birds, fish, and other aquatic creatures, including cumulative impacts to animals and populations impacted by numerous LNG facilities.

PHMSA has not always quantified or monetized safety benefits of its rulemakings,[96] as it is difficult to predict with any certainty how many safety incidents that would otherwise occur in

[96] *E.g., Pipeline Safety: Gas Pipeline Leak Detection and Repair*, 88 Fed. Reg. 31,890, 31,965 (May 18, 2023) (noting that quantified benefits of proposed leak detection rule include "the climate benefits of avoided methane emissions and the market value of avoided natural gas losses," while unquantified benefits include "safety

the baseline no action scenario a rulemaking may prevent, where such incidents would occur, and what impact such incidents could cause to what number of people. Even if PHMSA is unable to precisely monetize or exactly quantify such safety impacts, PHMSA must ensure that it qualitatively accounts for safety benefits—in the form of fewer lives lost, fewer injuries, less property damage, and the increased economic activity and access to recreation that is possible when people are safer and feel safer near LNG facilities—in both its NEPA review and its cost-benefit analysis of the LNG rulemaking.

Similarly, PHMSA must ensure that it adequately considers, whether quantitatively or qualitatively, other environmental and health impacts that may be difficult to monetize, such as benefits or harms related to air pollution, water pollution, light pollution, sound pollution, greenhouse gas emissions, and impacts to wildlife and their habitats.

Consideration of greenhouse gas emissions, in particular, is a necessary part of the required NEPA review because the substantive statute under which PHMSA is acting requires consideration of relevant environmental information, 49 U.S.C. § 60102(b)(2), as well as "a reasoned determination that the benefits, including safety and environmental benefits, of the intended standard justify its costs," *id.* § 60102(b)(5). PHMSA has previously considered the benefit of avoided GHG emissions in rulemakings. *See, e.g., Pipeline Safety: Safety of Gas Transmission Pipelines: Repair Criteria, Integrity Management Improvements, Cathodic Protection, Management of Change, and Other Related Amendments*, 87 Fed. Reg. 52,224, 52,264 (Aug. 24, 2022) (final rule considering "avoided environmental harms (including greenhouse gas emissions)" as a benefit of rule); *Pipeline Safety: Gas Pipeline Leak Detection*

---

benefits from early detection of leaks before they can evolve into incidents" and "environmental and public health benefits associating with preventing releases of natural gas, and other flammable, toxic, or corrosive gases").

and Repair, 88 Fed. Reg. 31,890, 31,893 (May 18, 2023) (proposed rule noting that a benefit of rulemaking would be to "reduce threats to the environment (including, but not limited to, reduction of methane emissions contributing to the climate crisis)"). Consistent with that past practice and its statutory duty to consider environmental costs and benefits of rulemaking, PHMSA must do so here as well.

VI. Conclusion

We urge PHMSA to initiate a thorough review and update of all of its LNG facility safety regulations that takes into account the changes in the industry, the unique and cumulative risks to human safety and the environment posed by large LNG export facilities, and the latest science that fills any previous knowledge gaps about topics such as vapor cloud explosions. Such a review is required in order to fulfill PHMSA's statutory duties to regulate for safety and environmental protection and specifically to update its LNG facilities standards as required by the PIPES Act of 2016 and the PIPES Act of 2020. Such a review is also necessary to develop standards that are as protective as possible to appropriately ensure the safety of Americans living near or working in the increasing number of geographically-concentrated LNG export facilities and to ensure protection of the environment. While there are many good reasons for PHMSA to move forward with incorporation of technical standards to include the most current version of those standards, as industry and GAO have urged PHMSA to do, this action alone would not be sufficient to ensure safety and environmental protection or to fulfill PHMSA's outstanding statutory mandates.

Respectfully submitted,

Ann Jaworski
Earthjustice
311 S. Wacker Drive, Suite 1400

Chicago, IL 60606
(773) 245-0837
ajaworski@earthjustice.org

Hana Vizcarra
Earthjustice
1001 G Street NW, Suite 1000
Washington, D.C. 20001
(202) 667-4500
hvizcarra@earthjustice.org

Elizabeth Livingston de Calderon
Earthjustice
900 Camp Street, unit 303
New Orleans, LA 70130
(504) 910-1712
ecalderon@earthjustice.org

Erin Murphy
Magdalen Sullivan
Environmental Defense Fund
555 12th St. NW, Suite 400
Washington, DC 20004
emurphy@edf.org
msullivan@edf.org